Curvature-adapted homogeneous hypersurfaces in symmetric spaces of non-compact type

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Curvature-adapted submanifolds

Let M be Riemannian manifold, F a submanifold, $p \in F$, $X \in T_pF$ a tangent vector and $\xi \in \nu_p F$ a normal vector.

- **Intrinsic curvature:** *Jacobi operator* $\mathcal{R}_\xi X = \mathcal{R}(X,\xi)\xi$.
- **Extrinsic curvature:** *Shape operator* $\mathcal{S}_\xi X = -(\nabla_X \xi)^\top$.

Both operators are self-adjoint, so they diagonalize for some basis.

Do they diagonalize simultaneously for some basis?

Equivalently: **Do they commute?**

Main definition

F is said to be *curvature-adapted to* M with respect to ξ at p if:

- $\mathcal{R}_\xi(T_pF) \subset T_pF$,
- $\mathcal{R}_\xi \circ \mathcal{S}_\xi = \mathcal{S}_\xi \circ \mathcal{R}_\xi$.

If this happens for every normal vector ξ at any point $p \in M$, we say that F is *curvature-adapted to* M [1].

Motivation and state of the art

Space forms ($\mathbb{R}^n$, $\mathbb{S}^n$ and $\mathbb{R}\mathrm{H}^n$) have constant sectional curvature, their Riemannian curvature is constant and their Jacobi operator is proportional to the identity.

Every submanifold in a space form is curvature-adapted but this is not true in general.

There are classifications of curvature-adapted hypersurfaces with *constant principal curvatures (cpc)*, i.e. whose shape operator $\mathcal{S}_\eta$ has constant eigenvalues for any unit normal vector η.

$\mathbb{C}\mathrm{P}^n$	$\mathbb{H}\mathrm{P}^n$ (without cpc)	$\mathbb{O}\mathrm{P}^2$
Kimura	Berndt	Murphy
Curvature-adapted + cpc $\Leftrightarrow$ Homogeneous		

$\mathbb{C}\mathrm{H}^n$	$\mathbb{H}\mathrm{H}^n$
Berndt	
Homogeneous $\not\Rightarrow$ Curvature-adapted	
Curvature-adapted + cpc $\Rightarrow$ Homogeneous	

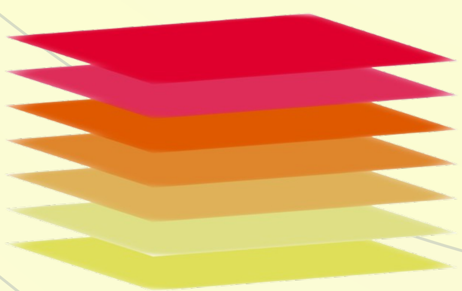
Objective

Investigate curvature-adapted hypersurfaces in symmetric spaces, especially assuming homogeneity, constant principal curvatures, or isoparametricity. In this poster we focus on **curvature-adapted homogeneous hypersurfaces** in symmetric spaces of non-compact type.

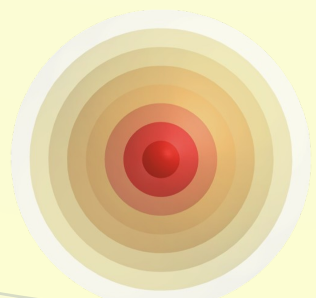
Key tool: Cohomogeneity one actions

Let H be a group of isometries of M. The isometric action $H \curvearrowright M$ has *cohomogeneity one (C1)* if it has orbits of codimension 1.

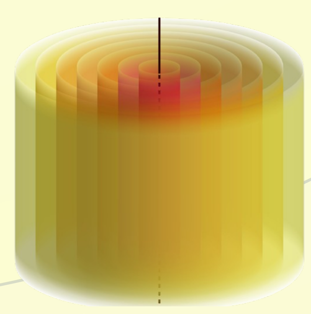
An orbit of codimension greater than 1 is called *singular*.



$\mathbb{R}^2 \curvearrowright \mathbb{R}^3$



$\mathrm{SO}(3) \curvearrowright \mathbb{R}^3$



$\mathrm{SO}(2) \times \mathbb{R} \curvearrowright \mathbb{R}^3$

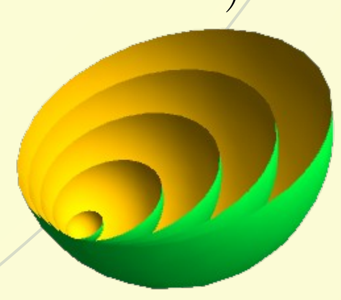
Classification of C1 actions

Let M be a symmetric space of non-compact type and $H \curvearrowright M$ a C1 action. Then, one of the following is satisfied [2]:

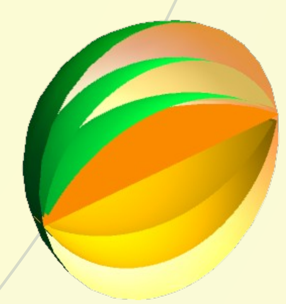
- The H-orbits form a foliation of horospherical or solvable type.
- $H \curvearrowright M$ has exactly one singular orbit that can be totally geodesic or not.

In case there is a non-totally geodesic singular orbit, the action can be constructed by one of these methods:

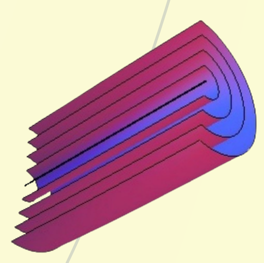
- The canonical extension.
- The nilpotent construction (only gives new examples in spaces of rank 1 and 2).



Horospherical foliation



Solvable foliation



C1 action with one singular orbit

The canonical extension method

Let M be a Riemannian manifold and H a subgroup of isometries of M whose action on M is **polar**. Hence, there is a totally geodesic submanifold Σ intersecting all the H-orbits orthogonally.

Assume further that the H-action on M is **free** and its orbits intersect Σ exactly **once**.

Let F be a submanifold of Σ and consider the set

$$H \cdot F = \{h \cdot p \in M : h \in H, p \in F\},$$

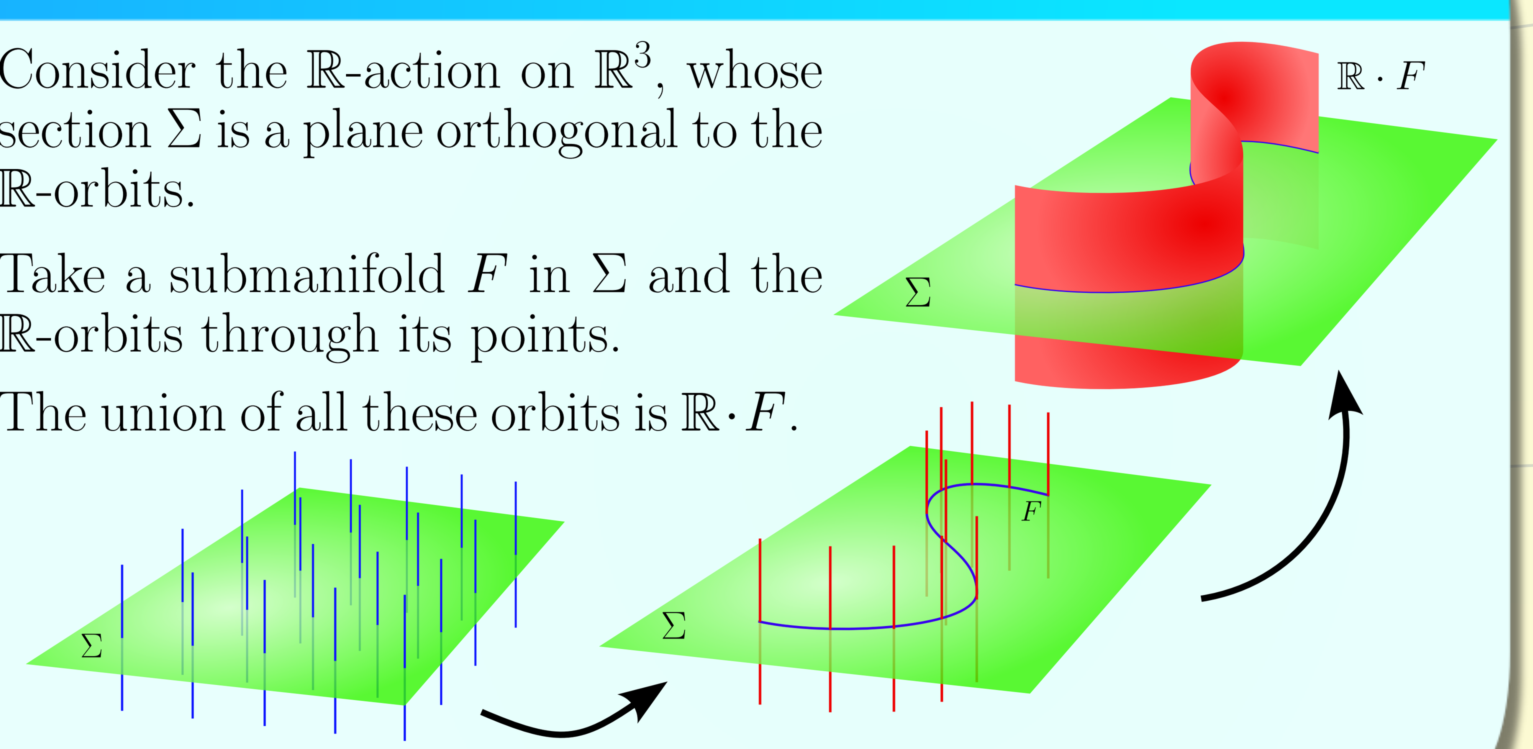
which turns out to be a submanifold in M.

Example

Consider the $\mathbb{R}$-action on $\mathbb{R}^3$, whose section Σ is a plane orthogonal to the $\mathbb{R}$-orbits.

Take a submanifold F in Σ and the $\mathbb{R}$-orbits through its points.

The union of all these orbits is $\mathbb{R} \cdot F$.



Proposition (R-V, SdT, S-L)

$H \cdot F$ is curvature-adapted to M if, and only if, F is curvature adapted to Σ and $H \cdot p$ is curvature-adapted to M with respect to any normal vector of F as a submanifold of Σ.

Main results

Proposition (R-V, SdT, S-L)

Let M be a symmetric space and F a totally geodesic submanifold. F is curvature-adapted to M if, and only if, it is the set of fixed points of an involutive isometry (it is reflective).

Proposition (R-V, SdT, S-L)

The nilpotent construction method does not give any new example of C1 action with curvature-adapted orbits.

Theorem (R-V, SdT, S-L)

Let M be an irreducible symmetric space of non-compact type and F a homogeneous curvature-adapted hypersurface of M. Then:

- F is a leaf of a horospherical foliation.
- F is a leaf of the canonical extension of a solvable foliation in $\mathbb{R}\mathrm{H}^n$,
- F is a tube about the canonical extension of a reflective submanifold.

References

[1] J. Berndt, L. Vanhecke, Curvature-adapted submanifolds, *Nihonkai Math J.* **3** (1992), no. 2, 177-185.
[2] V. Sanmartín, I. Solonenko, Classification of cohomogeneity-one actions on symmetric spaces of noncompact type, arXiv:2501.05553.